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Message from the ICSE 2021 General Chair
ICSE 2021

On behalf of the ICSE 2021 Organizing Committee, I'm delighted to welcome you to the 43rd ACM/IEEE International Conference on Software Engineering. For the first time in its 46-year history, ICSE is to visit Spain, although, unfortunately, it has to be virtually.

ICSE 2021 was originally scheduled as a physical event to take place in Madrid, Spain. The persistence of the coronavirus pandemic has, however, ruled out the possibility of holding a safe physical conference. It was with great sadness that we, therefore, decided to hold **ICSE 2021 and its co-located events and workshops in virtual format from May, 17 to June, 4.** This is obviously not what we had in mind five years ago when we worked hard to get ICSE to Madrid. We have put in a tremendous effort since on what would have been the first ICSE to be held in Spain. The entire team has redoubled its efforts to offer you an unforgettable virtual experience. The physical event that would give you the chance to enjoy Spanish food, music, dance and more will have to wait for better times.

Despite the enormous challenges that everybody is facing during these hard times, we are pleased that ICSE continues to be the premier software engineering conference. At the time of writing, the main conference had received over 1158 submissions which is in line with the figures for previous years. This year's main conference is one of the largest ever, with over 325 paper presentations. Paper presentations within the Technical, Software Engineering in Practice (SEIP), Software Engineering in Society (SEIS), New Ideas and Emerging Results (NIER) and Journal-First Tracks have been combined into blended sessions that are organized by topic rather than by track. The track on Software Engineering Education and Training (SEET) has its own topics and dedicated sessions. Technical briefings, as well as demos and posters will be run in parallel to the technical or networking sessions.

This year ICSE will include a four-day main conference taking place from Tuesday, May 25 to Friday, May 28. The two traditional mentoring events will each be held on two half-days prior to the main conference: the Doctoral Symposium (which provides students with constructive feedback on their PhD research) will run on May 21 and 24, while the New Faculty Symposium (which provides guidance to new faculty members on different facets of an academic career) will be held on May 19 and 20. The main conference will be preceded by five days of co-located conferences (May 17-21) and followed by 21 workshops dealing with specific aspects of software engineering spread over a seven-day period (May 29 to June 4).

The nine co-located events at ICSE 2021 are:

- **AST 2021 –** 2nd IEEE/ACM Conference on Automation of Software Test
- **FormaliSE 2021 –** 9th Conference on Formal Methods in Software Engineering
- **ICPC 2021** - 29th IEEE/ACM International Conference on Program Comprehension
- **ICSSP/ICGSE 2021 –** Joint 15th International Conference on Software and System Processes (ICSSP) and 16th ACM/IEEE International Conference on Global Software Engineering (ICGSE)
- **MOBILESoft 2021** – 8th ACM/IEEE International Conference on Mobile Software Engineering and Systems
- **MSR 2021** – 18th International Conference on Mining Software Repositories
- **SEAMS 2021** – 16th International Symposium on Software Engineering for Adaptive and Self-Managing Systems
- **TechDebt 2021 –** 4th International Conference on Technical Debt.

We have introduced a number of new initiatives this year regarding organization, tracks and virtual environment.

In terms of the organization, the innovations apply to:

1. A One Role Policy, deployed at ICSE 2021 in an attempt to reinforce diversity. According to this policy, nobody can participate in more than one committee, including the Organizing Committee. We would like to thank all the chairs for the big effort that they have made to roll out this policy. The results are promising, and committees (521 people) are more diverse and balanced than in previous years, including people from all continents (48% Europe; 30% North America; 11% Asia; 4.5% Oceania; 6% South America; 0.6% Africa) and a higher percentage of women (37% vs 64%).

2. This is the first ICSE to have Equity, Diversity and Inclusion (EDI) chairs. EDI chairs are in charge of promoting equity, diversity and inclusion at ICSE and ensuring that ICSE is perceived by attendees as being inclusive.

3. This is also the first ICSE to reckon with a Sustainability chair. The Sustainability chair is in charge of encouraging ICSE to promote all aspects of people- and planet-related sustainability.

4. Finally, ICSE 2021 has embraced open science as a strategic goal. Open science aims to promote the transparency and availability of research outputs at ICSE, encouraging better reproducibility and replicability of studies. This year we have required all authors of the main conference to upload their papers preprints. Such availability is critical in a virtual conference

Some tracks have implemented innovations, all of which are detailed in their respective welcome letters. We highlight some key innovations below:

1. The Technical Track set up the new role of area chairs to assist the program co-chairs. This year the Technical Track has a flat structure, without a separate program board from the Program Committee.

2. The Journal-First Track selected papers from submissions based on a number of criteria, including evident novelty with respect to already published papers, match with the established ICSE 2021 program, prospective interest to a large portion of the ICSE audience, and whether the authors were already presenting as part of one of the other tracks at the conference.

3. The Software Engineering Education and Training (SEET) Track has joined forces with the Conference on Software Engineering Education and Training (CSEE&T) to bring you the Joint Track on Software Engineering Education and Training (JSEET). This unification offers both communities a seamless, expanded, and richer experience.

Regarding the virtual environment:

1. We will again use the Clowdr platform, with which most of you are already familiar from ICSE 2020. Clowdr was hastily developed by our ICSE community as a solution for the ICSE 2020 virtualization issue in response to the COVID-19 pandemic. The Clowdr platform has grown and improved enormously. It is now a UK community interest company and, as such, is required to reinvest almost all of its revenue into improving the platform for the good of the academic community whence it originated. By opting for the Clowdr platform, not only are we supporting the future development of Clowdr but we are also promoting an affordable solution for all academic conference organizers in these difficult times.

2. The ICSE 2021 program has been divided into a 12-hour uniform daily schedule rather than using time zones. This should guarantee a simpler, uniform structure for ICSE 2021 that is easier for conference attendees to understand and transmit the feeling of a real conference. We chose the daily 12-hour slot that maximizes the synchronous participation of the ICSE community in order to promote high interactivity. We also provide mirroring to give attendees who could not attend the live program the chance to replay and watch the entire program during the following daily 12-hour slot.

3. We tried to improve networking activities and imitate the face-to-face experience. Although they are obviously no substitute for a physical conference, virtual environments can enable virtual networking with people all over the globe who would otherwise be unable to attend in person. Therefore, virtuality is an opportunity to increase access to the conference for many who, even under normal circumstances, may not have been able to travel to a physical conference.

4. We have made a big effort to keep up some ICSE traditions, even though the virtual context is not always helpful in this respect. We will be holding the usual ICSE run, albeit adapted to the virtual environment. We will be offering tidbits of Spanish culture: a live guitar player from the monumental ancient city of Caceres and an online Flamenco class and party.

5. We have made an effort to provide support for ICSE attendees and presenters to help them adapt to the virtual environment, offering tips to promote smooth attendance.

I am especially pleased to welcome five distinguished keynote speakers and encourage participants to attend their exciting plenary events.

- On Tuesday 14:00 CEST, Dario Gil, Senior Vice President and Director of IBM Research will introduce us to quantum computation and IBM Quantum Experience (platform that gives users access to a set of IBM's prototype quantum processors via the cloud). The talk discusses also about Qiskit challenges as error correction and verification. Qiskit provides tools for creating and manipulating quantum programs.

- On Tuesday 18:00 CEST, Prem Devanbu, Professor at UC Davis, will give an overview of naturalness and bimodality, as two important properties of software code, highlighting creative and interesting applications of naturalness, and the promise of bimodality.

- On Wednesday 10:00 CEST, Michael R. Lyu, Choh-Ming Li Professor at the Computer Science and Engineering Department, Chinese University of Hong Kong, will discuss the critical challenges commonly seen in industrial cloud systems and propose the development of a reliability-driven AIOps (Artificial Intelligence for IT Operations) framework to achieve resilient cloud systems.

- On Wednesday 17:30 CEST, Elaine Weyuker, University Distinguished Professor at the College of Engineering and Computer Science, University of Central Florida, will discuss her ideas on good software engineering research, and proper software engineering education based on her professional experience in different roles and positions in industry and academia.

- On Thursday 18:00 CEST, Jeannette Wing, Avanessians Director of the Data Science Institute and Professor of Computer Science at Columbia University, will invite us to ensure the responsible use of data to benefit society. In particular, she will present research activities aimed to promote Data for Good (the use of data to address societal challenges and bring in humanistic perspectives).

We have also scheduled three plenary sessions to honor the recipients of the three main ICSE awards, who will offer inspiring talks for the audience.

- ICSE 2011 Most Influential Paper award has gone to Andrea Arcuri and Lionel Briand for their paper entitled *A Practical Guide for Using Statistical Tests to Assess Randomized Algorithms in Software Engineering*, addressing the key challenges of using statistical tests in software engineering. We will enjoy their talk on Thursday 13:45 CEST.

- ACM SIGSOFT Outstanding Research award has gone to Prem Devanbu for profoundly changing the way researchers think about software by exploring connections between source code and natural language. Prem was invited as a keynote speaker months before he was honored with this award. Therefore, we have designed a special style for his two talks, combining both making longer his keynote on Tuesday.

- IEEE TCSE Harlan D. Mills award has gone to Dieter Rombach in recognition of his outstanding contributions to and leadership of research, teaching, and technology transfer in the area of empirical software engineering. He will delight us with his talk on Friday 13:45 CEST.

ICSE 2021 would not happen without the tremendous effort and countless hours of unpaid work put in by members of our community. Our heartfelt gratitude goes to everyone involved, and especially to our extraordinary Organizing Committee —a team of 71 volunteers drawn from Spain and around the world— that planned a physical conference and organized a virtual event: no easy feat.

All Organizing Committee members have done a splendid job, meeting deadlines and performing their roles with prowess. We have all worked a lot harder than we ever expected. You have no idea what it is to arrange such a complex conference as ICSE, the complications you face when the plans change half-way down the road as they did this year. The list of names is far too long to pen here, but they are omnipresent on ICSE 2021 website.

I cannot fail to draw attention to a few people who worked beyond the remit of their roles, laboring tirelessly to promptly address ICSE 2021 requirements. I am especially indebted to some members of the team who have provided support to me as General Chair, a role which at times involved an overwhelming workload: the Conference Chair, Oscar Dieste, our Project Manager, Claudia Ayala, and our Web Chair, Beatriz Bernández.

Finally, I would also like to thank all the authors and other contributors for participating in ICSE despite the difficult times that we are going through. You are the reason why ICSE is the premier conference in software engineering.

We are excited about the great program that we have prepared! I hope that you will find lots of interesting sessions and speakers in the technical program and plenty of opportunities for discussions and networking in the social program.

Before I finish, I would like to spare a thought for any members of the ICSE community that are suffering distress due to COVID-19. We wish you and your loved ones all the best in these difficult times.

Thank you,

Natalia Juristo
Universidad Politécnica de Madrid
General Chair

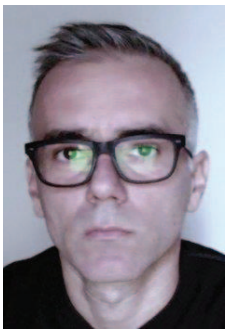
Message from the Demonstrations Track Chairs of ICSE 2021

ICSE-Companion 2021

Welcome to the ICSE 2021 Demonstrations Track. This track aims to inspire the software engineering community and advance research and practice with demonstrations of the latest software engineering tools. Researchers and practitioners present their tools in an interactive fashion, engaging in discussions with attendees and exchanging ideas and feedback.

We received 62 submissions this year. They went through a thorough review process, each paper being reviewed by at least 3 reviewers. All papers underwent an extensive online discussion, and we encouraged all reviewers to take a look at the outcome for all papers, to have a shared understanding of the results of the revision process. Finally, the program committee decided to accept 23 papers (37% acceptance rate).

We would like to thank the authors for their submissions, and the members of the Program Committee for their detailed reviews and careful discussions. We hope you enjoy the program and have inspiring interactions with authors and attendees at virtual ICSE 2021.



Giuseppe Scanniello, *University of Basilicata*
Demonstrations Co-Chair

Francisco Servant, *Virginia Tech*
Demonstrations Co-Chair

Demonstration – Program Committee
ICSE-Companion 2021

Iftekhar Ahmed, *University of California, Irvine*
Craig Anslow, *Victoria University of Wellington*
Venera Arnaoudova, *Washington State University*
Sebastian Baltes, *QAware GmbH and The University of Adelaide*
Amiangshu Bosu, *Wayne State University*
Sven Casteleyn, *Universidad Jaime I*
Jürgen Cito, *TU Wien and MIT*
Kostadin Damevski, *Virginia Commonwealth University*
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Martin Robillard, *McGill University*
Gema Rodriguez Perez, *University of Waterloo*
Simone Romano, *University of Bari*
Giuseppe Scanniello, *University of Basilicata*
Francisco Servant, *Virginia Tech*
Hitesh Sajnani, *Microsoft*
Sreedevi Sampath, *University of Maryland Baltimore County*
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Fiorella Zampetti, *University of Sannio, Italy*
Yongjie Zheng, *California State University San Marcos*

Demonstration – Additional Reviewers

ICSE-Companion 2021

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Carlos Bernal-Cárdenas, *William and Mary*
Maria Caulo, *University of Basilicata*
Nathan Cooper, *William & Mary*
Carmen Coviello, *University of Basilicata*
Sarah Fakhoury, *Washington State University*
David Nader Palacio, *William and Mary*
Rajshakhar Paul, *Wayne State University*
Devjeet Roy, *Washington State University*
Jaydeb Sarker, *Wayne State University*
Michael Schröder, *TU Wien*
Sayma Sultana, *Wayne State University*
Asif Kamal Turzo, *Wayne State University*
Ziyi Zhang, *University of Science and Technology of China*

Message from the Poster Track Chairs of ICSE 2021

ICSE-Companion 2021

Welcome to the ICSE 2021 Posters Track, which provides a prime opportunity for researchers and practitioners to present, showcase, and discuss their most recent advances, ideas, experiences, and challenges in the field of software engineering at the premier software engineering venue. The main purpose of the ICSE 2021 Posters Track is to promote discussion of grand scientific challenges, future visions, potentially disruptive ideas, and recent advances towards cutting-edge research results with the broad scientific community attending the conference.

As in past editions, the ICSE 2021 Posters Track received submissions only on an invitational basis without an open call. Work submitted to the ICSE 2021 Technical Track have been invited by the Chairs to submit poster papers (i.e., 2-page extended abstracts). As a consequence, the innovative aspect of poster submissions was ensured by the Technical Track's thorough review process. The 2-page extended abstracts of poster submissions still had to properly summarize (1) the problem addressed by the poster, (2) the approach taken by the authors, (3) and the main results achieved so far.

Each of the poster papers submitted to the ICSE 2021 Posters Track was reviewed by three members of the program committee in a single-blinded review process. Based on the reviews, six poster submissions have been accepted for presentation at Virtual ICSE 2021 fostering discussion on recent advances on multi-language software, program translation, bug fixing, consistency checking of engineering artefacts, metamorphic testing and test sample generation.

We would like to thank all people who contributed to the ICSE 2021 Posters Track. A special thanks goes to the authors and the members of the Program Committee, who had to review poster papers within a short timeframe. We would like to thank Natalia Juristo, the General Chair of ICSE 2021 for her invaluable advice and guidance in organizing this track.

We would also like to express our gratitude to Claudia Ayala, Project Manager of ICSE 2021, Ignacio Panach, Proceedings Co-Chair, and Beatriz Bernárdez, Web Chair of ICSE 2021, for making smooth and efficient the organisation of this track.

We are also grateful for the ideas and hints exchanged with Arie van Deursen and Tao Xie, Program Chairs of ICSE 2021, and Sebastian Proksch, Virtualization Chairs of ICSE 2021.

On behalf of the entire Program Committee, we would like to welcome all attendees of the Virtual ICSE 2021 Posters Track and wish them fruitful discussions.

Davide Fucci, *Blekinge Institute of Technology, Sweden*
Posters Track Co-Chair

Federica Sarro, *University College London, UK*
Posters Track Co-Chair

Posters – Program Committee
ICSE-Companion 2021

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Giovanni Denaro, *University of Milano-Bicocca, Italy*
Davide Fucci, *Blekinge Institute of Technology*
Gregory Gay, *Chalmers and the University of Gothenburg*
Michael W. Godfrey, *University of Waterloo, Canada*
Carmine Gravino, *University of Salerno*
Lucas Gren, *Chalmers, University of Gothenburg and Volvo Cars*
Giovani Guizzo, *University College London*
Maria Kechagia, *University College London*
Rahul Mohanani, *fortiss GmbH*
Tien N. Nguyen, *University of Texas at Dallas*
Elda Paja, *IT University of Copenhagen*
Mike Papadakis, *University of Luxembourg, Luxembourg*
Chaiyong Ragkhitwetsagul, *Mahidol University, Thailand*
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Silvia Regina Vergilio, *Federal University of Paranà*
Antonio Vetrò, *Politecnico di Torino*
Jéssyka Vilela, *Universidade Federal de Pernambuco*
Krzysztof Wnuk, *Blekinge Institute of Technology*

Message from the Doctoral Symposium Chairs of ICSE 2021

ICSE-Companion 2021

The ICSE 2012 Doctoral Symposium was a two-day event held on May 21 and May 24, 2021 virtually (originally in Madrid, Spain). The Doctoral Symposium provided an opportunity for doctoral students to present their ongoing work and future research plans, and to discuss them with experienced researchers and fellow doctoral students in the Software Engineering community. Students participating in the Doctoral Symposium presented their work to a panel of experts who provided each student with valuable feedback on their work and highlighted opportunities to further improve their research work. In addition, the symposium facilitated the exchange of ideas among young researchers.

Twenty-nine students made submissions to the Doctoral Symposium; twenty-one corresponded to students in an early stage of their PhD; and eight were in a late stage of their PhD. Submissions underwent a thorough review with each submission receiving feedback from three expert reviewers. Following extensive discussion amongst the reviewers and chairs, nine early submissions and six late submissions were accepted. A three-page summary of each early PhD paper and a six-page summary of each late PhD paper appears in these proceedings.

Every doctoral student of a late PhD paper performed a formal presentation at the Doctoral Symposium, while students with early PhD papers were divided in three parallel working groups where they presented their research. In all cases, they were given feedback by the Doctoral Panel.

The topics presented at the ICSE Doctoral Symposium give us a glimpse into some topics we may see in the future of software engineering research. The participants work on various topics relevant to software engineering including software product lines, open source software, software security, software testing, program comprehension and several other exciting topics. Everyone attending the Doctoral Symposium benefited from the presentations and discussions of this important contemporary work from our up-and-coming new software engineering researchers.

To offer students practical guidance on completion of their research and initiation of future research career, the symposium also included an informal panel on “Performing Doctoral Studies” and plenty of opportunities for social interactions.

We would like to express our gratitude to the Doctoral Panel for providing extensive and constructive feedback to the doctoral students and for their participation during the event, and all doctoral students who submitted a proposal to the Doctoral Symposium. We would also like to thank the ICSE 2021 organization committee for helping us deal with all the logistics involved in running the event.

Sebastian Uchitel, *Universidad de Buenos Aires and Imperial College London*
Doctoral Symposium Co-Chair

Sira Vegas, *Universidad Politecnica de Madrid*
Doctoral Symposium Co-Chair

Doctoral Symposium – Program Committee
ICSE-Companion 2021

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Tevfik Bultan, *University of California, Santa Barbara*
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Jane Cleland-Huang, *University of Notre Dame*
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Stephen MacDonell, *University of Otago*
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David Rosenblum, *National University of Singapore*
Jocelyn Simmonds, *University of Chile*
Sebastian Uchitel, *Universidad de Buenos Aires / Imperial College*
Sira Vegas, *Universidad Politecnica de Madrid*
Claes Wohlin, *Blekinge Institute of Technology*
Chang Xu, *Nanjing University*

Message from the Student Research Competition Chairs of ICSE 2021

ICSE-Companion 2021

Following the tradition of past years, ICSE 2021 is hosting the ACM Student Research Competition (SRC), sponsored by Microsoft Research. The competition offers undergraduate and graduate students a unique forum to experience the research world, present their research results to conference attendees, get direct feedback on their work from a panel of expert judges, meet other students, and compete for prizes. The winners of the competition are invited to participate in the ACM Student Research Competition Grand Finals, where they will compete against winners from other conferences held during 2021.

Participants in the SRC can be undergraduate or graduate students currently pursuing a master or PhD. To enroll in the competition, a student must submit a research abstract (two pages) related to any of the main ICSE themes. Submissions have been reviewed by the SRC Program Committee members and those that are accepted to the competition are published in the ICSE conference companion proceedings. In the 2021 edition, the SRC has received 35 submissions, 7 from undergraduate students and 28 from graduate students. After the evaluation of each abstract by three PC members, 5 undergraduate and 16 graduate students were invited to compete at the ACM SRC at ICSE 2021.

During the conference dates, students of accepted abstracts will present a poster describing their work, which will be evaluated by a panel of judges during one of the two planned sessions. In this second round, the judges will evaluate students' knowledge of their research area, the contribution of their research, and the quality of their oral and visual presentation. The best qualified students from each category will undergo the third round of the competition, which consists in a final presentation followed by questions from the judges during a dedicated session. Three undergraduate and three graduate students will be chosen as winners and will receive awards during the closing ceremony.

Organizing a student research competition is a great team effort, especially this year that ICSE is a virtual event. We would like to thank all the committee members and judges for their willingness and commitment to participate in ICSE SRC 2021. It has been a pleasure to work with such a brilliant team and coordinate this important ICSE 2021 event. Also, we want to express our gratitude to the ICSE organization committee, the General Chair Natalia Juristo, and the virtualization team led by Sebastian Proksch, for the support in the organization of the competition live sessions. We also thank ACM and Microsoft for supporting this event. Finally, we wish good luck to all the students and we hope they take the most of this enriching experience at ICSE 2021.

Sergio Segura, *University of Seville, Spain*
SRC Co-Chair

Alessandro Garcia, *PUC-Rio, Brazil*
SRC Co-Chair

Aurora Ramirez, *University of Córdoba, Spain*
SRC Co-Chairs

Student Research Competition Committee of ICSE 2021

ICSE-Companion 2021

Andrew Bergel, *Microsoft*
Tsong Yueh Chen, *Swinburne University of Technology*
Francisco Chicano, *University of Málaga*
J. Andres Diaz-Pace, *ISISTAN Research Institute and UNICEN University*
Alastair Donaldson, *Imperial College London*
Tracy Hall, *Lancaster University*
Kim Herzig, *Microsoft*
Robert M. Hierons, *The University of Sheffield*
Upulee Kanewala, *University of North Florida*
Miqing Li, *University of Birmingham*
Ana Paiva, *University of Porto*
Rick Rabiser, *Johannes Kepler University*
Awais Rashid, *University of Bristol*
Alejandro Russo, *Chalmers University of Technology*

Student Research Competition – Program Committee

ICSE-Companion 2021

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Tsong Yueh Chen, *Swinburne University of Technology*
Francisco Chicano, *University of Málaga*
Andres Diaz Pace, *UNICEN University*
Alastair Donaldson, *Imperial College London*
Tracy Hall, *Lancaster University*
Kim Herzig, *Tools for Software Engineers*
Rob Hierons, *The University of Sheffield*
Upulee Kanewala, *Montana State University*
Miqing Li, *University of Birmingham*
Ana Paiva, *Faculty of Engineering of the University of Porto*
Rick Rabiser, *LIT CPS*
Aurora Ramírez, *University of Córdoba*
Awais Rashid, *University of Bristol*
Alejandro Russo, *Chalmers University of Technology*
Sergio Segura, *Universidad de Sevilla*
Gustavo Soares, *Microsoft*
Shin Hwei Tan, *Southern University of Science and Technology*
Javier Troya, *Universidad de Málaga*
Junbeom Yoo, *Konkuk University*

Message from the Artifact Evaluation Chairs of ICSE 2021

ICSE-Companion 2021

Artifacts are digital objects created by researchers as part of their studies or generated by their experiments. In the field of software engineering artifacts, can be software systems and scripts used to run experiments, input datasets, raw data collected in the experiment, or scripts used to analyze results.

High-quality artifacts of published research papers increase the likelihood that research results can be independently replicated and reproduced by other researchers. Experience also shows that papers providing artifacts earn increased citations and greater prestige in the research community.

In this spirit, the artifact evaluation track aims to promote, share, and catalog the research artifacts of software engineering papers. Authors of papers accepted to the technical track were invited to submit their artifact for evaluation as a candidate for reusable, available, replicated or reproduced artifact. In addition and as in previous year, authors of any prior SE work (published at ICSE or elsewhere) were also invited to submit an artifact for evaluation as a candidate replicated or reproduced artifact.

We received 59 submissions. Each artifact was reviewed by at least two members of our evaluation committee and further assessed in the online discussion phase. If satisfying the criteria, artifacts were awarded the respective badge, which are shown on the front page of their paper and on the conference website. This year, 12 artifacts were finally accepted as *Reusable* and 39 as *Available* (which was rewarded to artifacts which were also reusable). In addition, as a further form of recognition — and for the first time at ICSE — all accepted extended abstracts documenting the artifacts were published in the ICSE Companion Volume.

We would like to express our gratitude to the authors for carefully preparing and submitting research artifacts. We would also like to thank our evaluation committee for providing constructive feedback and actively helping the authors in further improving their artifacts. It has been a privilege working with the 38 members of our evaluation committee! Finally, we would also like to thank the ICSE 2021 organization committee for helping us deal with all the logistics involved in organizing the artifact evaluation track.

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Daniel Mendez, *Blekinge Institute of Technology, Sweden*
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Message from the ICSE 2021 Technical Briefings Chairs

ICSE-Companion 2021

Software Engineering is a constantly changing field, with areas continuously appearing or evolving in response to emerging technologies and challenges. This perpetual motion creates challenges for researchers and practitioners in the field, who may not have the time and resources that are needed for learning the new methods, techniques, models, and languages.

The ICSE Technical Briefings track addresses this challenge. Technical Briefings provide conference participants exposure to the latest advances in the field, taught by world-leading researchers and practitioners. They offer a unique opportunity to the community to get acquainted with different software engineering topics in an efficient and proficient manner.

The Technical Briefings track at ICSE 2021 solicited three types of submissions:

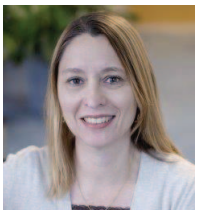
- General Technical Briefings, covering the general state-of-the-art in a wider area of software engineering (e.g., search-based software engineering)
- Specific Technical Briefings, offering a focused introduction to an emerging topic in software engineering (e.g., software engineering practices for specific types of systems or emerging technologies)
- Hands-on Technical Briefings, a tool-supported session on a specific topic with hands-on participation from attendees

Submissions were evaluated by three members of the program committee who took int consideration the type of submission when applying the evaluation criteria. We selected seven submissions for presentation in the conference:

- Two General Technical Briefings on Bayesian Data Analysis and Code Summarization.
- Two Specific Technical Briefings on Natural Language Processing (in the context of Requirements Engineering) and Grounded Theory.
- Three Hands-on Technical Briefings on Development of Trustworthy AI Software, Challenges in Building SmartChatbots, and Reactive Synthesis with the Spectra language and toolset.

We are confident that this final offering covers a wide range of interests for both software engineering researchers and practitioners.

Xavier Franch, *Polytechnical University of Catalunya, Barcelona*
Technical Briefings Co-Chair



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Antonio Vallecillo, *University of Málaga*
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Semantic Patches for Adaptation of JavaScript Programs to Evolving Libraries

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I. Purpose

The artifact contains the JSFIX tool along with instructions for how to replicate all of the experiments from the paper. The purpose of the artifact is to allow the the reader to check that the results presented in the paper are correct.

II. Requirements

The only requirement is some basic bash skills. The user should know how to open a terminal, and how to navigate the file system with e.g., the 'cd' command. All other commands can be copy pasted from the README file.

III. Hardware/software requirements

A computer running either Linux, Mac or Windows with at least 10GB of RAM that can be dedicated to the virtual machine and around 100GB of free disk space.

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